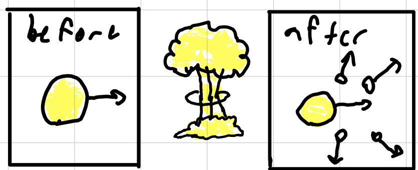
Tree Generation Idea

Conservation of Momentum

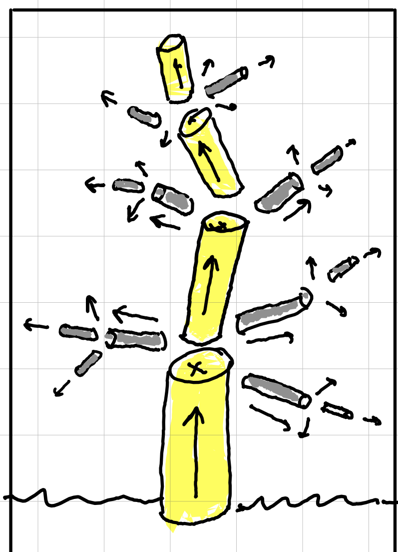
$\vec{p} = m\vec{v}$, $m = \frac{4}{3}\pi r^3$

$\sum \vec{p}_{initial} = \sum \vec{p}_{final}$

Explosion



General Routine



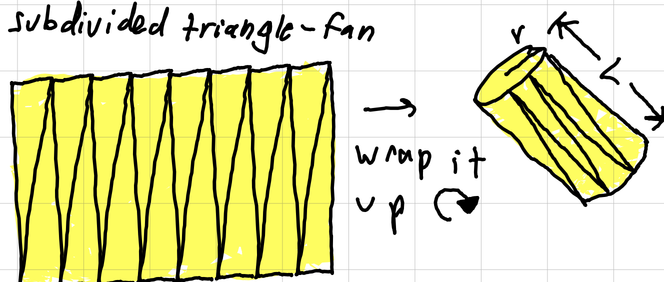
1. per branch

- Starting radius 

- mostly up direction  (θ near 90°)
- random extrude length L
- starting speed $\vec{v}$

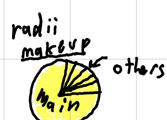
2. extrude cylinder for each branch

subdivided triangle-fan



3. after extrusion (explosion)

- Split into N pieces
- main branch gets 75-95% radius
- the other branches momentum totals the pre-explosion
- as well as energy conservation



$KE = \frac{1}{2}mv^2$

4. leaf Placement

- with gravity consideration 
- after N branches deep?